# AMR MOHAMED

**AI Engineer | Computer Vision & Full-Stack Engineer**
Cairo, Egypt | +20 100 270 0369 | amrcsai11@gmail.com
linkedin.com/in/amr-mohamed-a5b00924b | github.com/Elm3lmAmr

## PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Artificial Intelligence (Nile University, 2026) who builds and ships production AI systems end to end. At Edara Property (SODIC), delivered four enterprise platforms covering real-time face recognition, Arabic OCR, WebRTC body-cam streaming and AI delivery management. Skilled in computer vision, deep learning, Python/FastAPI, Node.js/NestJS, React, PostgreSQL and Docker, with a strong focus on security, data privacy and measurable outcomes.

## TECHNICAL SKILLS

**AI & Computer Vision:** Face Detection & Recognition (InsightFace, SCRFD, ArcFace), OCR (PaddleOCR, Arabic/RTL), OpenCV, ONNX Runtime, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CNN, LSTM, VAE, Diffusion Models, Reinforcement Learning, XGBoost

**Languages:** Python, C++, JavaScript, TypeScript, R, SQL, Java, HTML/CSS

**Backend:** FastAPI, NestJS, Node.js, Express, Django REST Framework, REST APIs, WebSockets, Socket.io, Server-Sent Events, WebRTC, JWT, RBAC

**Frontend & Mobile:** React, Vite, TailwindCSS, Flutter

**Data & DevOps:** PostgreSQL, MySQL, Prisma, TypeORM, Pydantic, Docker, Docker Compose, GitHub Actions (CI/CD), Traefik, nginx, S3/MinIO, MLflow, Git

**Testing & Security:** pytest, Jest, Vitest, AES-256-GCM Encryption, SHA-256 Integrity Checks, Helmet, Rate Limiting, Zod Validation

## PROFESSIONAL EXPERIENCE

**Edara Property Management (SODIC)** **July 2026 – Present**
*AI Engineer | Cairo, Egypt*

- Design, build and deploy production AI and full-stack systems for security operations, identity verification and internal delivery management across SODIC residential compounds.
- Delivered 4 enterprise platforms end to end, from requirements and system architecture to containerised production deployment.
- Applied security-by-design principles: encryption at rest, role-based access control, audit trails and alignment with Egypt's Personal Data Protection Law (151/2020).

**Mind Booster Academy** **Aug 2024 – July 2026**
*Programming Instructor*

- Designed and taught courses in Python, C++, Web Development, Data Analysis, Power BI and Scratch for learners aged 7 to 60.
- Built project-based curricula and mentored students in coding fundamentals and competitive programming.

## KEY PROJECTS – EDARA PROPERTY (SODIC)

**STMC – AI Face Recognition & Community Security Platform** **Sept 2026 – Present**
*Python, FastAPI, InsightFace, ONNX Runtime, OpenCV, NestJS, PostgreSQL, React, Docker, Traefik | Live:* isop.edaraproperty.net

- Built a real-time face recognition pipeline (SCRFD detection + ArcFace embeddings on ONNX Runtime) with face quality checks, cross-frame tracking and confirmed/tentative/unknown matching.
- Integrated multi-brand DVR/NVR cameras over RTSP (Hikvision, Dahua, Uniview, TVT) with brand detection and automatic stream reconnection.
- Re-architected the live camera wall to one shared server-side stream per camera, removing the browser's 4-camera limit and enabling unlimited simultaneous feeds.
- Built mobile self-enrollment with guided 3-pose face capture validated by AI head-pose and quality scores, plus owner/tenant flows and a security approval workflow.
- Secured biometric data with AES-256-GCM encryption at rest, RBAC and escalating login lockout; added enrollment funnel analytics to identify and reduce user drop-off.

**Egyptian National ID OCR – Automated Data Extraction System** **Sept 2026 – Present**
*Python, PaddleOCR, OpenCV, FastAPI, Pydantic, pytest*

- Built an on-device OCR pipeline that converts ID card photos into validated JSON, running fully on CPU with no cloud services and no data written to disk.
- Developed a custom Arabic digit classifier (glyph segmentation + template matching) achieving 100% accuracy on the 14-digit national ID number.

- Corrected PaddleOCR's reversed Arabic/RTL output and dropped zeros; implemented mod-11 checksum validation with auto-correction and derived date of birth, gender and governorate.
- Designed confidence-based routing (auto-accept / review / reject) that auto-processed 66% of a real production batch (150 of 226 cards), a 50% improvement over the previous version.
- Exposed the system as a REST API (FastAPI/Swagger) and batch CLI; built a synthetic data generator, evaluation harness and 40 unit tests.

**SSOP – Security Operations Platform (Body-Cam Live Streaming)** **Aug 2026 – Present**

*Flutter, Node.js, Express, WebRTC, WebSocket, MySQL, React, Leaflet*

- Turned guards' smartphones into body cameras streaming live 1080p/30fps video to a control room with under 500 ms latency via WebRTC and a custom WebSocket signaling server.
- Built background recording in an Android foreground service that survives screen lock, with separate incident clips and automatic end-of-shift upload.
- Delivered SOS and "Red Flag" alerts, SLA-tracked incident management, live GPS tracking and tamper-proof digital shift handovers.
- Ensured evidence chain of custody with SHA-256 fingerprinting, full access logging and PDF audit export; enforced RBAC across 6 user roles.

**NEXUS – Digital & AI Delivery Hub** **Aug 2026 – Present**

*TypeScript, React, Node.js, Express, PostgreSQL, Prisma, Socket.io, GitHub Actions*

- Built an internal platform managing the full delivery lifecycle: request intake, triage, task assignment, UAT, training with QR attendance and executive analytics.
- Modelled 18 database entities with Prisma migrations; built a derived-state engine and transactional triage that guarantee consistent request status.
- Implemented JWT authentication with refresh-token rotation, RBAC across 5 roles, real-time Socket.io notifications and direct-to-S3 presigned uploads.
- Delivered executive dashboards (SLA, cycle time, UAT pass rate, team workload) and automated CI/CD with GitHub Actions.

## ADDITIONAL PROJECTS

**AI Anti-Cheat System for FPS Games | AI Lead** **Jan 2025 – Present**

- Built 10+ ML/DL models (CNN, LSTM, VAE, Reinforcement Learning) to detect cheating behaviour; owned data preprocessing, feature engineering, training and evaluation. Code: github.com/Elm3lmAmr/AI-AntiCheat

**Avance Men Salon – Booking Web Application | Freelance Developer** **Mar 2024 – Present**

- Developed a full-stack booking website with service selection and real-time scheduling; managed client requirements and deployment. Live: avancebarbiere.com

**Real-Time Shipment Tracking Application | Backend Lead** **Aug 2023 – Dec 2023**

- Built Django REST APIs for live geolocation tracking and optimised SQL queries to reduce latency; integrated with a Flutter frontend.

## EDUCATION

**Bachelor of Science in Artificial Intelligence** **Graduated: July 2026**

*Nile University, Egypt | GPA: 3.0/4.0*

**Relevant Coursework:** Machine Learning, Deep Learning, NLP, Neural Networks, MLOps, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Database Systems

## ACHIEVEMENTS

- **Codeforces Pupil:** Max rating 1231 (codeforces.com/profile/elm3lmmr).
- **ECPC Participant:** Egyptian Collegiate Programming Contest 2023, 2024 and 2025.

## LANGUAGES

Arabic (Native) | English (Professional Working Proficiency, B2)